# RAJALAKSHMI ENGINEERING COLLEGE

**RAJALAKSHMI NAGAR, THANDALAM – 602 105**




## CS23331 - DESIGN AND ANALYSIS OF ALGORITHMS

**Laboratory Record Note Book**

Name :…….. ……...

Year / Branch / Section : …….. …….

University Register No. : …….. ……

College Roll No. : ……..……

Semester : ……..……

Academic Year : …….

# INDEX PAGE

| Exp.No | Name of the Experiment | Date of Completion |
|---|---|---|
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |

# Problem 1: Finding Complexity using Counter Method

| | |
|---|---|
| **Started on** | Wednesday, 5 August 2026, 9:58 AM |
| **State** | Finished |
| **Completed on** | Wednesday, 5 August 2026, 10:04 AM |
| **Time taken** | 6 mins 28 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**
A positive Integer n

**Output:**
Print the value of the counter variable

**For example:**

| Input | Result |
|---|---|
| 9 | 12 |

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
int c=0;
int i=1;
c++;
int s=1;
c++;
int n;
scanf("%d",&n);
while(s<=n){
c++;
i++;
c++;
s+=i;
c++;
}
c++;
printf("%d",c);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 9 | 12 | 12 | ✔ |
| ✔ | 4 | 9 | 9 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# Problem 2: Finding Complexity using Counter method

| | |
|---|---|
| **Started on** | Wednesday, 5 August 2026, 10:04 AM |
| **State** | Finished |
| **Completed on** | Wednesday, 5 August 2026, 10:25 AM |
| **Time taken** | 20 mins 29 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n,c=0;
    scanf("%d",&n);
    if(n==1){
        c++;
        printf("*");
    }
    else{
        for(int i=1;i<=n;i++){
            c++;
            for(int j=0;j<n;j++){
                c++;

              // printf("*");
             //  printf("*");
              c++;
                break;
            }
            c++;
            c++;
        }
       c++;
       c++;
    }

    printf("%d",c);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 2 | 12 | 12 | ✔ |
| ✔ | 1000 | 5002 | 5002 | ✔ |
| ✔ | 143 | 717 | 717 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# Problem 3: Finding Complexity using Counter Method

| | |
|---|---|
| **Started on** | Wednesday, 5 August 2026, 10:28 AM |
| **State** | Finished |
| **Completed on** | Wednesday, 5 August 2026, 10:32 AM |
| **Time taken** | 3 mins 47 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```c
#include<stdio.h>
int  main(){
    int num,c=0;
    scanf("%d",&num);
    for(int i=1;i<=num;i++){
        c++;
        if(num%i==0){
            c++;
            //printf("%d",i);
        }
        c++;
    }
    c++;
    printf("%d",c);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 31 | 31 | ✔ |
| ✔ | 25 | 54 | 54 | ✔ |
| ✔ | 4 | 12 | 12 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# Problem 4: Finding Complexity using Counter Method

| | |
|---|---|
| **Started on** | Wednesday, 5 August 2026, 10:33 AM |
| **State** | Finished |
| **Completed on** | Wednesday, 5 August 2026, 11:14 AM |
| **Time taken** | 40 mins 50 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include<stdio.h>
int main(){
    int n,c=0;
    scanf("%d",&n);
    c++;
    for(int i=n/2;i<n;i++){
       c++;

        for(int j=1;j<n;j=2*j){
            c++;

            for(int k=1;k<n;k=k*2){
                c++;

                c++;
            }
            c++;


        }
       c++;


    }
    c++;
    printf("%d",c);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4 | 30 | 30 | ✔ |
| ✔ | 10 | 212 | 212 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# Problem 5: Finding Complexity using counter method

| | |
|---|---|
| **Started on** | Wednesday, 5 August 2026, 11:15 AM |
| **State** | Finished |
| **Completed on** | Wednesday, 5 August 2026, 11:24 AM |
| **Time taken** | 9 mins 25 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**
A positive Integer n

**Output:**
Print the value of the counter variable

**Answer:**

```c
#include<stdio.h>
int main(){
    int n;
    int rev=0,rem,c=0;
    c++;
    c++;
    scanf("%d",&n);
    while(n!=0){

        rem=n%10;
        c++;
        rev=rev*10+rem;
        c++;
        n/=10;
        c++;
        c++;
    }
    c++;
    printf("%d",c);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 11 | 11 | ✔ |
| ✔ | 1234 | 19 | 19 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# 1-G-Coin Problem

| | |
|---|---|
| **Started on** | Wednesday, 26 August 2026, 8:04 PM |
| **State** | Finished |
| **Completed on** | Wednesday, 26 August 2026, 8:23 PM |
| **Time taken** | 19 mins 17 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

Write a program to take value V and we want to make change for V Rs, and we have infinite supply of each of the denominations in Indian currency, i.e., we have infinite supply of { 1, 2, 5, 10, 20, 50, 100, 500, 1000} valued coins/notes, what is the minimum number of coins and/or notes needed to make the change.

Input Format:

Take an integer from stdin.

Output Format:

print the integer which is change of the number.

Example Input :

64

Output:

4

Explanaton:

We need a 50 Rs note and a 10 Rs note and two 2 rupee coins.

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[9]={1000,500,100,50,20,10,5,2,1};
    int c=0;
    for(int i=0;i<9;i++){
        while(n>=a[i]){
            n-=a[i];
            c++;
        }
        if(n==0){
            break;
        }
    }
    printf("%d",c);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 49 | 5 | 5 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

# 2-G-Cookies Problem

| | |
|---|---|
| **Started on** | Wednesday, 26 August 2026, 8:24 PM |
| **State** | Finished |
| **Completed on** | Wednesday, 26 August 2026, 8:53 PM |
| **Time taken** | 29 mins 28 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

Assume you are an awesome parent and want to give your children some cookies. But, you should give each child at most one cookie.

Each child i has a greed factor g[i], which is the minimum size of a cookie that the child will be content with; and each cookie j has a size s[j]. If s[j] >= g[i], we can assign the cookie j to the child i, and the child i will be content. Your goal is to maximize the number of your content children and output the maximum number.

**Example 1:**

**Input:**

3

1 2 3

2

1 1

**Output:**

1

Explanation: You have 3 children and 2 cookies. The greed factors of 3 children are 1, 2, 3.

And even though you have 2 cookies, since their size is both 1, you could only make the child whose greed factor is 1 content.

You need to output 1.

**Constraints:**

1 <= g.length <= 3 * 10^4

0 <= s.length <= 3 * 10^4

1 <= g[i], s[j] <= 2^31 - 1

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
void sort(int a[],int n){
    for(int i=0;i<n-1;i++){
        for(int j=i+1;j<n;j++){
            if(a[i]>a[j]){
                int temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
        }
    }
}
int main(){
    int x,y;
    scanf("%d",&x);
    int g[x];
    for(int i=0;i<x;i++){
        scanf("%d",&g[i]);
    }
    scanf("%d",&y);
    int s[y];
    for(int i=0;i<y;i++){
        scanf("%d",&s[i]);
    }
    sort(g,x);
    sort(s,y);
    int i=0,j=0,c=0;
    while(i<x && j<y){
        if(s[i]>=g[i]){
            c++;
            i++;
            j++;
        }
        else{
            j++;
        }
    }
    printf("%d",c);

}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 2<br>1 2<br>3<br>1 2 3 | 2 | 2 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# 3-G-Burger Problem

| | |
|---|---|
| **Started on** | Monday, 28 September 2026, 9:25 PM |
| **State** | Finished |
| **Completed on** | Monday, 28 September 2026, 9:25 PM |
| **Time taken** | 30 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the perso
If he has eaten $i$ burgers with $c$ calories each, then he has to run at least $3^i * c$ kilometers to burn out
burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are $(3^0 * 1) + (3$
But this is not the minimum, so need to try out other orders of consumption and choose the minimum value.
he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm.Apply greedy

**Input Format**

First Line contains the number of burgers

Second line contains calories of each burger which is n space-separate integers

**Output Format**

Print: Minimum number of kilometers needed to run to burn out the calories

**Sample Input**

```
3
5 10 7
```

**Sample Output**

```
76
```

---

**For example:**

| Test | Input | Result |
|---|---|---|
| Test Case 1 | 3<br>1 3 2 | 18 |

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
#include<math.h>
#include<stdlib.h>

int cmp(const void *a, const void *b) {
    return (*(int*)b - *(int*)a);
}

int main(){
    int n,cnt=0;
    scanf("%d",&n);
    int d[n];
    for(int i=0;i<n;i++){
        scanf("%d",&d[i]);
    }
    qsort(d,n,sizeof(int),cmp);
    for(int i=0;i<n;i++){
        cnt += pow(n,i) * d[i];
    }
    printf("%d",cnt);
    return 0;
}
```

| | Test | Input | Expected | Got | |
|---|---|---|---|---|---|
| ✔ | Test Case 1 | 3<br>1 3 2 | 18 | 18 | ✔ |
| ✔ | Test Case 2 | 4<br>7 4 9 6 | 389 | 389 | ✔ |
| ✔ | Test Case 3 | 3<br>5 10 7 | 76 | 76 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# 4-G-Array Sum max problem

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 7:26 PM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 7:37 PM |
| **Time taken** | 11 mins 21 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(int i=0;i<n-1;i++){
        for(int j=i+1;j<n;j++){
            if(a[j]<a[i]){
                int temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
        }
    }
    int sum=0;
    for(int i=0;i<n;i++){
        sum+=a[i]*i;
    }
    printf("%d",sum);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>2<br>5<br>3<br>4<br>0 | 40 | 40 | ✔ |
| ✔ | 10<br>2<br>2<br>2<br>4<br>4<br>3<br>3<br>5<br>5<br>5 | 191 | 191 | ✔ |
| ✔ | 2<br>45<br>3 | 45 | 45 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# 5-G-Product of Array elements-Minimum

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 7:38 PM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 7:51 PM |
| **Time taken** | 12 mins 54 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct | Mark 1.00 out of 1.00 | Flag question

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs( 1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

**For example:**

| Input | Result |
|---|---|
| 3<br>1<br>2<br>3<br>4<br>5<br>6 | 28 |

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n],b[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
     for(int i=0;i<n;i++){
        scanf("%d",&b[i]);
    }
    for(int i=0;i<n-1;i++){
        for(int j=i+1;j<n;j++){
            if(a[i]>a[j]){
                int temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
        }
    }
    for(int i=0;i<n-1;i++){
        for(int j=i+1;j<n;j++){
            if(b[j]>b[i]){
                int temp=b[i];
                b[i]=b[j];
                b[j]=temp;
            }
        }
    }
    int sum=0;
    for(int i=0;i<n;i++){
        sum+=a[i]*b[i];
    }
    printf("%d",sum);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1<br>2<br>3<br>4<br>5<br>6 | 28 | 28 | ✔ |
| ✔ | 4<br>7<br>5<br>1<br>2<br>1<br>3<br>4<br>1 | 22 | 22 | ✔ |
| ✔ | 5<br>20<br>10<br>30<br>10<br>40<br>8<br>9<br>4<br>3<br>10 | 590 | 590 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# 1-Number of Zeros in a Given Array

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 7:54 PM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 8:14 PM |
| **Time taken** | 19 mins 40 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    int low=0,high=n-1;
    int firstzero=n;
    while(low<=high){
        int mid=(low+high)/2;
        if(a[mid]==1){
            low=mid+1;
        }
        else{
            firstzero=mid;
            high=mid-1;
        }
    }
    printf("%d",n-firstzero);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |
| ✔ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 | 0 | 0 | ✔ |
| ✔ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0 | 8 | 8 | ✔ |
| ✔ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

# 2-Majority Element

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 8:19 PM |
| **State** | Finished |
| **Completed on** | Saturday, 29 August 2026, 10:37 PM |
| **Time taken** | 2 days 2 hours |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

**Example 1:**

```
Input: nums = [3,2,3]
Output: 3
```

**Example 2:**

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

**Constraints:**

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

**For example:**

| Input | Result |
|---|---|
| 3<br>3 2 3 | 3 |
| 7<br>2 2 1 1 1 2 2 | 2 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int count(int a[], int l, int r, int value) {
    int c = 0;
    for (int i = l; i <= r; i++) {
        if (a[i] == value)
            c++;
    }
    return c;
}

int majority(int a[], int l, int r) {
    if (l == r)
        return a[l];
    int mid = (l+ r) / 2;
    int x = majority(a, l, mid);
    int y = majority(a, mid + 1, r);
    if (x == y)
        return x;
    int cx = count(a, l, r, x);
    int cy = count(a, l, r, y);

    if (cx > cy)
        return x;
    else
        return y;
}

int main() {
    int n;
    scanf("%d", &n);
    int a[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);
    printf("%d", majority(a, 0, n - 1));
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>3 2 3 | 3 | 3 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# 3-Finding Floor Value

| | |
|---|---|
| **Started on** | Saturday, 29 August 2026, 10:37 PM |
| **State** | Finished |
| **Completed on** | Saturday, 29 August 2026, 10:44 PM |
| **Time taken** | 6 mins 59 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

**Problem Statement:**
Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

**Input Format**
First Line Contains Integer n – Size of array
Next n lines Contains n numbers – Elements of an array
Last Line Contains Integer x – Value for x

**Output Format**
First Line Contains Integer – Floor value for x

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int main() {
    int n;
    scanf("%d", &n);
    int a[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
    int x;
    scanf("%d", &x);
    int low = 0;
    int high = n - 1;
    int floor = -1;
    while (low <= high) {
        int mid = (low + high) / 2;
        if (a[mid] <= x) {
            floor = a[mid];
            low = mid + 1;
        }
        else {
            high = mid - 1;
        }
    }
    printf("%d", floor);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 6<br>1<br>2<br>8<br>10<br>12<br>19<br>5 | 2 | 2 | ✔ |
| ✔ | 5<br>10<br>22<br>85<br>108<br>129<br>100 | 85 | 85 | ✔ |
| ✔ | 7<br>3<br>5<br>7<br>9<br>11<br>13<br>15<br>10 | 9 | 9 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# 4-Two Elements sum to x

| | |
|---|---|
| **Started on** | Saturday, 29 August 2026, 10:44 PM |
| **State** | Finished |
| **Completed on** | Saturday, 29 August 2026, 11:04 PM |
| **Time taken** | 20 mins |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int findPair(int a[], int left, int right, int x, int *p, int *q) {
    if (left >= right) {
        return 0;
    }
    int mid = (left + right) / 2;
    if (findPair(a, left, mid, x, p, q)) {
        return 1;
    }
    if (findPair(a, mid + 1, right, x, p, q)) {
        return 1;
    }
    for (int i = left; i <= mid; i++) {
        for (int j = mid + 1; j <= right; j++) {
            if (a[i] + a[j] == x) {
                *p = a[i];
                *q = a[j];
                return 1;
            }
        }
    }

    return 0;
}

int main() {
    int n;
    scanf("%d", &n);
    int a[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
    int x;
    scanf("%d", &x);
    int p, q;
    if (findPair(a, 0, n - 1, x, &p, &q)) {
        printf("%d\n", p);
        printf("%d\n", q);
    } else {
        printf("No");
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4<br>2<br>4<br>8<br>10<br>14 | 4<br>10 | 4<br>10 | ✔ |
| ✔ | 5<br>2<br>4<br>6<br>8<br>10<br>100 | No | No | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# 5-Implementation of Quick Sort

| | |
|---|---|
| **Started on** | Saturday, 29 August 2026, 11:05 PM |
| **State** | Finished |
| **Completed on** | Saturday, 29 August 2026, 11:08 PM |
| **Time taken** | 3 mins 29 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

**For example:**

| Input | Result |
|---|---|
| 5<br>67 34 12 98 78 | 12 34 67 78 98 |

**Answer:**

```c
#include <stdio.h>
void quickSort(int a[], int low, int high) {
    if (low < high) {
        int pivot = a[high];
        int i = low - 1;
        for (int j = low; j < high; j++) {
            if (a[j] < pivot) {
                i++;
                int temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
        int temp = a[i + 1];
        a[i + 1] = a[high];
        a[high] = temp;
        int p = i + 1;
        quickSort(a, low, p - 1);
        quickSort(a, p + 1, high);
    }
}

int main() {
    int n;
    scanf("%d", &n);
    int a[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
    quickSort(a, 0, n - 1);
    for (int i = 0; i < n; i++) {
        printf("%d ", a[i]);
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>67 34 12 98 78 | 12 34 67 78 98 | 12 34 67 78 98 | ✔ |
| ✔ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✔ |
| ✔ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# 1-DP-Playing with Numbers

| | |
|---|---|
| **Started on** | Friday, 2 October 2026, 6:44 PM |
| **State** | Finished |
| **Completed on** | Friday, 2 October 2026, 6:58 PM |
| **Time taken** | 14 mins 22 secs |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** Correct Mark 10.00 out of 10.00 Flag question

**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

**Example 1:**

***Input:*** *6*
***Output:****6*
***Explanation:*** *There are 6 ways to 6 represent number with 1 and 3*

*1+1+1+1+1+1*
*3+3*
*1+1+1+3*
*1+1+3+1*
*1+3+1+1*
*3+1+1+1*

**Input Format**

First Line contains the number n

**Output Format**

**Print: The number of possible ways 'n' can be represented using 1 and 3**

Sample Input

6

Sample Output

6

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int a;
    scanf("%d",&a);
    long long d[a+1];
    d[0]=1;
    for(int i=1;i<=a;i++){
        d[i]=d[i-1];
        if(i>=3){
            d[i]+=d[i-3];
        }
    }
    printf("%lld",d[a]);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 6 | 6 | 6 | ✔ |
| ✔ | 25 | 8641 | 8641 | ✔ |
| ✔ | 100 | 24382819596721629 | 24382819596721629 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Finish review

Back to Course

# 2-DP-Playing with chessboard

| | |
|---|---|
| **Started on** | Friday, 2 October 2026, 6:59 PM |
| **State** | Finished |
| **Completed on** | Friday, 2 October 2026, 7:20 PM |
| **Time taken** | 21 mins 13 secs |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 10.00 out of 10.00 Flag question

**Playing with Chessboard:**

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

**Example:**

**Input**

3
**1** 2 4
**2** 3 4
**8 7 1**

**Output:**

19

**Explanation:**

Totally there will be 6 paths among that the optimal is
Optimal path value:1+2+8+7+1=19

**Input Format**

First Line contains the integer n
The next n lines contain the n*n chessboard values

**Output Format**

Print Maximum monetary value of the path

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n][n];
    int d[n][n];
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            scanf("%d",&a[i][j]);
        }
    }
    d[0][0]=a[0][0];
    for(int i=1;i<n;i++){
        d[0][i]=a[0][i]+d[0][i-1];
    }
    for(int i=1;i<n;i++){
        d[i][0]=a[i][0]+d[i-1][0];
    }
    for(int i=1;i<n;i++){
        for(int j=1;j<n;j++){
            if(d[i-1][j]>d[i][j-1]){
                d[i][j]=d[i-1][j]+a[i][j];
            }
            else{
                d[i][j]=d[i][j-1]+a[i][j];
            }
        }
    }
    printf("%d",d[n-1][n-1]);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 2 4<br>2 3 4<br>8 7 1 | 19 | 19 | ✔ |
| ✔ | 3<br>1 3 1<br>1 5 1<br>4 2 1 | 12 | 12 | ✔ |
| ✔ | 4<br>1 1 3 4<br>1 5 7 8<br>2 3 4 6<br>1 6 9 0 | 28 | 28 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Finish review

Back to Course

# 3-DP-Longest Common Subsequence

| | |
|---|---|
| **Started on** | Friday, 2 October 2026, 7:35 PM |
| **State** | Finished |
| **Completed on** | Friday, 2 October 2026, 7:47 PM |
| **Time taken** | 12 mins 23 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct | Mark 1.00 out of 1.00 | Flag question

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| s1 | a | g | **g** | **t** | **a** | **b** | |
| s2 | **g** | x | **t** | **x** | **a** | **y** | **b** |

**The length is 4**

Solveing it using Dynamic Programming

**For example:**

| Input | Result |
|---|---|
| aab<br>azb | 2 |

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
#include<string.h>
int main(){
    char s1[100],s2[100];
    scanf("%s\n%s",s1,s2);
    int a=strlen(s1);
    int b=strlen(s2);
    char d[a+1][b+1];
    for(int i=0;i<=a;i++){
        d[i][0]=0;
    }
    for(int j=0;j<=b;j++){
        d[0][j]=0;
    }
    for(int i=1;i<=a;i++){
        for(int j=1;j<=b;j++){
            if(s1[i-1]==s2[j-1]){
                d[i][j]=d[i-1][j-1]+1;
            }
            else{
                if(d[i-1][j]>d[i][j-1]){
                    d[i][j]=d[i-1][j];
                }
                else{
                    d[i][j]=d[i][j-1];
                }
            }
        }
    }
    printf("%d",d[a][b]);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | aab<br>azb | 2 | 2 | ✔ |
| ✔ | ABCD<br>ABCD | 4 | 4 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# 4-DP-Longest non-decreasing Subsequence

| | |
|---|---|
| **Started on** | Friday, 2 October 2026, 7:52 PM |
| **State** | Finished |
| **Completed on** | Friday, 2 October 2026, 8:01 PM |
| **Time taken** | 8 mins 56 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    int d[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
        d[i]=1;
    }

    for(int i=1;i<n;i++){
        for(int j=0;j<i;j++){
            if(a[i]>=a[j]){
                if(d[j]+1>d[i]){
                    d[i]=d[j]+1;
                }
            }
        }
    }
    int max=d[0];
    for(int i=1;i<n;i++){
        if(d[i]>max){
            max=d[i];
        }
    }
    printf("%d",max);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 9<br>-1 3 4 5 2 2 2 2 3 | 6 | 6 | ✔ |
| ✔ | 7<br>1 2 2 4 5 7 6 | 6 | 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

# 1-Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity

| | |
|---|---|
| **Started on** | Wednesday, 7 October 2026, 9:44 AM |
| **State** | Finished |
| **Completed on** | Wednesday, 7 October 2026, 9:52 AM |
| **Time taken** | 7 mins 56 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int  n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(int i=0;i<n-1;i++){
        int c=0;
        int x=a[i];
        for(int j=i+1;j<n;j++){
            if(x==a[j]){
                c++;
                a[j]=-1;
            }
        }
        if(c!=0){
            printf("%d",x);
        }
    }
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# 2-Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

| | |
|---|---|
| **Started on** | Wednesday, 7 October 2026, 10:06 AM |
| **State** | Finished |
| **Completed on** | Wednesday, 7 October 2026, 10:28 AM |
| **Time taken** | 21 mins 53 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    int count[n+1];
    for(int i=0;i<=n;i++){
        count[i]=0;
    }
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
        count[a[i]]++;
    }
    for(int i=0;i<n+1;i++){
        if(count[i]>1){
            printf("%d",i);
            break;
        }
    }
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# 3-Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Complexity

| | |
|---|---|
| **Started on** | Friday, 11 September 2026, 3:26 PM |
| **State** | Finished |
| **Completed on** | Friday, 11 September 2026, 3:26 PM |
| **Time taken** | 25 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
|---|---|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

void solve() {
    int n1;
    if (scanf("%d", &n1) != 1) return;
    int arr1[n1];
    for (int i = 0; i < n1; i++) {
        scanf("%d", &arr1[i]);
    }

    int n2;
    scanf("%d", &n2);
    int arr2[n2];
    for (int i = 0; i < n2; i++) {
        scanf("%d", &arr2[i]);
    }

    int i = 0, j = 0;
    int first = 1;
    while (i < n1 && j < n2) {
        if (arr1[i] < arr2[j]) {
            i++;
```

**For example:**

| Input | Result |
| --- | --- |
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

void solve() {
    int n1;
    if (scanf("%d", &n1) != 1) return;
    int arr1[n1];
    for (int i = 0; i < n1; i++) {
        scanf("%d", &arr1[i]);
    }

    int n2;
    scanf("%d", &n2);
    int arr2[n2];
    for (int i = 0; i < n2; i++) {
        scanf("%d", &arr2[i]);
    }

    int i = 0, j = 0;
    int first = 1;
    while (i < n1 && j < n2) {
        if (arr1[i] < arr2[j]) {
            i++;
        } else if (arr1[i] > arr2[j]) {
            j++;
        } else {
            if (!first) {
                printf(" ");
            }
            printf("%d", arr1[i]);
            first = 0;
            i++;
            j++;
        }
    }
    printf("\n");
}

int main() {
    int t;
    if (scanf("%d", &t) == 1) {
        while (t--) {
            solve();
        }
    }
    return 0;
}
```

| | Input | Expected | Got | |
| --- | --- | --- | --- | --- |
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

# 4-Print Intersection of 2 sorted arrays-O(m+n)Time Complexity,O(1) Space Complexity

| | |
|---|---|
| **Started on** | Wednesday, 7 October 2026, 10:32 AM |
| **State** | Finished |
| **Completed on** | Wednesday, 7 October 2026, 10:33 AM |
| **Time taken** | 1 min |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct | Mark 1.00 out of 1.00 | Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
|---|---|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int n, m;
        scanf("%d", &n);

        int a[n];
        for (int i = 0; i < n; i++)
            scanf("%d", &a[i]);

        scanf("%d", &m);

        int b[m];
        for (int i = 0; i < m; i++)
            scanf("%d", &b[i]);

        int i = 0, j = 0;

        while (i < n && j < m) {
            if (a[i] == b[j]) {
                printf("%d ", a[i]);
                i++;
                j++;
            }
            else if (a[i] < b[j]) {
                i++;
            }
            else {
                j++;
            }
        }

        printf("\n");
    }

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

# 5-Pair with Difference-O(n^2)Time Complexity,O(1) Space Complexity

| | |
|---|---|
| **Started on** | Wednesday, 7 October 2026, 10:33 AM |
| **State** | Finished |
| **Completed on** | Wednesday, 7 October 2026, 10:34 AM |
| **Time taken** | 57 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

**For example:**

| Input | Result |
|---|---|
| 3<br>1 3 5<br>4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, k;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &k);

    int i = 0, j = 1;

    while (j < n) {
        int diff = a[j] - a[i];

        if (diff == k && i != j) {
            printf("1");
            return 0;
        }
        else if (diff < k) {
            j++;
        }
        else {
            i++;
        }

        if (i == j)
            j++;
    }

    printf("0");

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 3 5<br>4 | 1 | 1 | ✔ |
| ✔ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1 | 1 | 1 | ✔ |
| ✔ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0 | 0 | 0 | ✔ |
| ✔ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

# 6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

| | |
|---|---|
| **Started on** | Wednesday, 7 October 2026, 10:34 AM |
| **State** | Finished |
| **Completed on** | Wednesday, 7 October 2026, 10:35 AM |
| **Time taken** | 42 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

**For example:**

| Input | Result |
|---|---|
| 3<br>1 3 5<br>4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, k;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &k);

    int i = 0, j = 1;

    while (j < n) {
        if (a[j] - a[i] == k) {
            printf("1");
            return 0;
        }
        else if (a[j] - a[i] < k) {
            j++;
        }
        else {
            i++;
        }

        if (i == j)
            j++;
    }

    printf("0");

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 3 5<br>4 | 1 | 1 | ✔ |
| ✔ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1 | 1 | 1 | ✔ |
| ✔ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0 | 0 | 0 | ✔ |
| ✔ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course